

Histology Notes Histology of Circulatory System

General Structure of Blood Vessels

Most blood vessels have 3 tunics (layers):

1. Tunica Intima (Interna)

▼ Innermost layer facing lumen.

Made of:

- ✓ Simple squamous epithelium (endothelium)
- ✓ Basement membrane
- ✓ Subendothelial loose connective tissue
- ✓ Internal elastic lamina (mainly arteries)

2. Tunica Media

▼ Middle layer.

Contains:

- ✓ Smooth muscle cells
- ✓ Elastic fibers
- ✓ Reticular fibers
- 🎯 Controls vessel diameter and blood pressure.

3. Tunica Adventitia (Externa)

▼ Outer connective tissue layer.

Contains:

- ✓ Collagen fibers
- ✓ Elastic fibers
- ✓ Nerves
- ✓ Vasa vasorum (in large vessels)



Histology of Heart

The heart wall has 3 layers:

1 Endocardium

▼ Equivalent to vascular tunica intima.

Structure:

- ✓ Simple squamous endothelial cells
- ✓ Thin connective tissue layer
- ✓ Elastic + collagen fibers
- ✓ Some smooth muscle cells

Subendocardial layer contains:

- ✓ Blood vessels
- ✓ Nerves
- ✓ Purkinje fibers

2 Myocardium

▼ Thickest layer of heart wall

Contains:

- ✓ Cardiac muscle cells
- ✓ Conducting system cells

Thickness:

Ventricles > atria

Left ventricle is thickest because it pumps systemic circulation.

♦ Purkinje Fibers ⚡

- ✓ Specialized cardiac muscle cells.
- ✓ Located in subendocardial region.
- ✓ Conduct electrical impulses through the heart.

3 Epicardium

- ✓ Outer layer of heart.
- ✓ Equivalent to visceral layer of serous pericardium.

Covered by:

- ✓ Mesothelium (simple squamous epithelium)

Contains:

- ✓ Loose connective tissue
- ✓ Fat cells
- ✓ Coronary vessels
- ✓ Nerves

🌟 Function:

Produces lubricating fluid → reduces friction during heart movement.

Cardiac Valves

Mainly composed of:

- ✓ Dense connective tissue
- ✓ Covered by endothelium
- ✓ Connected to fibrous skeleton of heart.
- ✓ Chordae tendineae anchor valve leaflets.

Arteries

General Features

- ✓ Carry blood away from heart
- ✓ Thick walls
- ✓ Lumen smaller compared with veins.

Types:

Elastic Arteries (Conducting Arteries)

Examples:

- ▼ Aorta
- ▼ Pulmonary artery

Features:

- ▼ Largest arteries (>10 mm)
- ▼ Yellow color due to elastin.

Tunica Intima:

- 🎯 Thick
- 🎯 Endothelium
- 🎯 Thick subendothelial connective tissue

Tunica Media:

Most important feature:

- ✓ Many elastic lamellae
- ✓ Smooth muscle layers

Function:

- ✓ Stretch during systole
- ✓ Recoil during diastole
- ✓ Maintains continuous blood flow

Adventitia:

Contains:

- ✓ Vasa vasorum
- ✓ Sympathetic nerves

2 Muscular Arteries (Distributing Arteries)

Examples:

- ✓ Femoral artery
- ✓ Brachial artery

Function:

- ✓ Distribute blood to organs.

Histology:

Intima:

- ✓ Endothelium
- ✓ Subendothelial CT
- ✓ Prominent internal elastic lamina

Media:

Main identifying feature:

- ✓ Thick smooth muscle layer

Adventitia:

- ✓ Collagen fibers
- ✓ Elastic fibers
- ✓ Nerves
- ✓ Lymphatics

Small Arteries

Diameter:

✓ 0.1–2 mm

Features:

✓ Thin wall

✓ Narrow lumen

Media:

✓ Up to 8 layers smooth muscle

✓ No external elastic lamina

Arterial Sensory Receptors

Carotid Sinus ★

Function:

✓ Detects blood pressure changes.

Contains:

✓ Baroreceptors

✓ Sensory fibers from CN IX (glossopharyngeal nerve)

Response:

↑ BP → stretch receptor activation → brain response → vasodilation ↓ BP

Glomus Bodies

▼ Chemoreceptors.

▼ Detect blood chemistry:

👉 Oxygen

👉 Carbon dioxide

👉 pH

CAPILLARIES

General Features

✓ Smallest blood vessels

✓ Site of exchange of gases, nutrients & wastes

✓ Wall consists only of endothelium + basement membrane

1 Continuous Capillaries

Features:

- 👉 No pores (fenestrae)
- 👉 Continuous basal lamina

📌 Found in:

Muscle

Connective tissue

CNS

Exocrine glands

2 Fenestrated Capillaries

Features:

- 👉 Have pores (fenestrae)
- 👉 Continuous basal lamina
- 👉 Rapid exchange

📌 Found in:

Kidney

Intestine

Endocrine glands

Pancreas

3 Sinusoidal Capillaries

Features:

- 👉 Large irregular lumen
- 👉 Large gaps between cells
- 👉 Discontinuous basal lamina

📌 Found in:

Liver

Spleen

Bone marrow

Placenta

VEINS

General Features

- ✓ Carry blood toward the heart
- ✓ Thin walls
- ✓ Large lumen
- ✓ Often contain valves

Types of Veins

1 Venules

- Smallest veins

Functions:

- 👉 Drain capillaries
- 👉 Site of leukocyte migration during inflammation 

2 Small Veins

- 👉 Thin intima
- 👉 Well-developed adventitia

3 Medium Veins

- Femoral vein, cephalic vein

Features:

- 👉 Thin media
- 👉 Prominent adventitia
- 👉 Valves present

4 Large Veins

- Superior vena cava (SVC)
- Inferior vena cava (IVC)
- Portal vein

Features:

- 👉 Thick adventitia
- 👉 Thin media